

OTEPIC

COMPANION PLANTING



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TRAINING MANUAL HANDBOOK

COMPANION PLANTING

The practice of growing different plant species together for mutual benefit, offers several advantages to farmers and gardeners.

Companion planting is a gardening technique that involves planting different crops in close proximity to each other to benefit from their mutual relationship. Companion planting is an important part of Perma gardening, which is a permanent, bio-intensive garden that produces a variety of healthy foods.

A garden mini-ecosystem is part of a larger ecosystem, interacting with sun, shade, warmth, wind, birds, insects, and animals. Nature's ecosystem is varied and balanced, with harmonious, beneficial relationships. While our main focus is on growing a healthy soil, an additional goal is to make our garden reflect nature's diversity. Even weeds have a key role to play. Companion planting involves choosing which crops to put beside each other for the best results, keeping in mind the garden as a whole.



BENEFICIAL INFLUENCES

Following are some plants that have been found by experience to be good for gardens in general. They are perennial, so they might be planted at the end of the beds where they will not get in the way of double digging. Some are herbs that can be enjoyed for tea or seasoning; others are weeds that we would do well to encourage in our gardens instead of trying to get rid of:

Lemon balm (tea)	Oregano (herb)	Dandelion Chamomile
(tea)	Marjoram (herb)	Stinging nettle Valerian (root is medical)

Here are the key benefits:

1. **Pest Control:** Natural Pest Repellence: Certain plants can naturally repel pests that target other crops. For example, marigolds can deter nematodes, while basil can help keep aphids away from tomatoes.
2. **Attracting Beneficial Insects:** Companion plants can attract beneficial insects such as pollinators (bees, butterflies) and natural pest predators (ladybugs, lacewings), helping control harmful pests naturally.
3. **Disruption of Pest Lifecycles:** By planting complementary species, pests may find it more difficult to locate their preferred host plants, reducing infestations.
4. **Improved Plant Growth:** Nutrient Sharing: Some plants can fix nitrogen in the soil (e.g., legumes like beans) and provide nutrients to neighboring plants that require more nitrogen. This can reduce the need for chemical fertilizers.
5. **Better Use of Space:** Companion planting can maximize space efficiency by pairing plants with different growth habits. For instance, tall plants like corn can provide shade for shorter crops like lettuce.
6. **Enhanced Growth through Symbiosis:** Certain plants, such as corn, beans, and squash (known as the “Three Sisters” planting), support each other by providing physical support, soil cover, and improved soil health.
7. **Improved Pollination:** Attracting Pollinators: Companion plants can attract a variety of pollinators, which can enhance fruit and seed production in neighboring plants. Flowers like sunflowers or lavender can draw in bees, improving pollination for nearby crops like tomatoes or cucumbers.

8. **Weed Suppression:** Smothering Weeds: Fast-growing plants or ground covers, such as clover or squash, can shade the soil and prevent the growth of weeds by blocking sunlight.
9. **Competitive Growth:** Certain plants can outcompete weeds for nutrients, water, and light, reducing the need for herbicides.
10. **Improved Soil Health:** Soil Enrichment: Companion plants that fix nitrogen, such as beans, peas, or clover, help enrich the soil, benefiting neighboring plants that require nitrogen. This can reduce the need for synthetic fertilizers.
11. **Root Interaction:** Different root systems from companion plants can help improve soil structure, prevent soil compaction, and promote better water infiltration.
12. **Enhanced Crop Yields:** Increased Efficiency: By strategically planting complementary crops, you can optimize the use of available resources (light, water, nutrients), often leading to higher yields.
13. **Reduced Crop Failure:** Companion planting can increase biodiversity in the garden or field, reducing the likelihood of complete crop failure due to pests, diseases, or weather extremes.
14. **Reduced Reliance on Chemicals:** Natural Pest and Disease Management: Companion planting reduces the need for chemical pesticides and herbicides by encouraging natural pest control and plant resilience.
15. **Fewer Soil Depleting Inputs:** By improving soil health naturally through companion planting, you reduce the need for chemical fertilizers and synthetic soil amendments.
16. **Flavor Enhancement:** Improved Taste: Some plants, such as basil and tomatoes, are believed to improve each other's flavor when grown together. This can enhance the overall quality and taste of the produce.
17. **Aesthetic and Aesthetic Diversity:** Visual Appeal: Companion planting can create more aesthetically pleasing gardens or farms by incorporating a variety of colors, textures, and plant shapes. This can enhance the beauty of the growing space while also supporting ecological balance.
18. **Cultural and Culinary Benefits:** Growing diverse plant species together can support cultural traditions and culinary preferences, allowing for the cultivation of unique plant combinations.

19. **Promoting Biodiversity:** Supporting Ecosystem Health: Companion planting encourages biodiversity, which improves the health of the ecosystem by creating habitats for a variety of species, from insects to soil organisms.
20. **Reduces Disease Spread:** Biodiverse systems can reduce the spread of diseases by limiting the monoculture environment that pests and pathogens thrive in.



SUN/SHADE

It is easy to forget that tiny seedlings that can turn into tall plants such as corn can be put where it will shade a plant that enjoys less sun, like peas or potatoes or cucumbers. Sun-loving tomatoes can provide a cooler mini- climate for onions or parsley. Notice that the potatoes in the lazy bed will be shaded by the corn, and the tomatoes, will shade the onions, if the bed is oriented as indicated. Cool weather crops, like lettuce, carrots, onions, and potatoes, will do well in partial shade in warmer weather.



ATTRACTING “GOOD BUGS”

Bees and butterflies play an important part in the life cycle of plants, so a garden will benefit if it includes their favorite meals. Bees can account for one-third of the United States crop yield through pollination they accomplish. Bees love blue flowers, especially Borage and Rosemary. Butterflies are attracted to purple, red, yellow, and orange flowers and will beautify our garden along with the flowers you plant to attract them. Other beneficial insects are attracted to the flowers of parsley, dill and cilantro/coriander. Try letting a few of those plants go to seed to serve as feeding stations for helpful insects.

KEEPING THE GARDEN HEALTHY

Beginning gardeners are often inclined to worry about getting rid of insects and weeds, but it is much more enjoyable to think of insects and weeds as part of nature’s contribution to a diverse ecosystem. Weeds that compete with the plants we are trying to grow should obviously be taken out and added to the compost pile, and insects that insist on helping themselves to our garden

vegetables need to be related with gently yet firmly. Generally, a garden will benefit from the gardeners focus on health and life rather than on death and disease. A thriving diverse garden with healthy soil will attract beneficial insects that will make themselves useful pollinating, cleaning up rotting debris, and eating harmful insect larvae. In fact, in a balanced mini-ecosystem, for every seven or eight good bugs, there will be only one harmful one. If we get rid of all the bad bugs in our garden, the good bugs have less to eat and have no reason to stay around to help. Insects and disease are most likely to attack sick plants, those that are under stress for some reason.



Making sure the soil has all the nutrients, soil air, soil moisture and cured compost needed by plants we are growing is a much better way to use our energy than looking for ways to get rid of pests. Compost made from a variety of plant materials will encourage a variety of microorganisms in the soil, and they will provide the wide range of nutrients and microorganisms needed for healthy plants. Careful transplanting also helps to promote uninterrupted root growth and encourages vigorous, healthy plants. The right amount of water throughout the plants growing period will also reduce the likelihood of stress.

Other Benefits of Companion Planting

- **Pest control:** Some plants can confuse pests, while others attract beneficial insects that prey on pests. For example, nasturtiums can deter aphids from beans, and radishes can repel flea beetles.
- **Pollination:** Some plants can improve pollination of fruit and vegetable crops.

- **Weed suppression:** Some plants can act as a natural mulch to block sunlight and reduce weed growth.
- **Nutrient exchange:** Some plants can provide nutrients to other plants. For example, beans provide nitrogen to the soil.
- **Space maximization:** Companion planting can help maximize the use of space.

Some examples of companion plant combinations include:

- **The Three Sisters:** A popular combination of corn, beans, and squash. The corn provides structure for the beans to climb, the beans provide nitrogen for the other plants, and the squash creates a living mulch.
- **Nasturtiums and beans:** Nasturtiums can deter aphids from beans.
- **Alliums and carrots:** Alliums can ward off carrot root fly.
- **Pumpkins and green beans:** Pumpkin leaves can act as a natural mulch and weed control for beans.

Conclusion

Companion planting is a sustainable agricultural practice that provides a wide range of benefits, from improved pest control and plant health to higher yields and reduced environmental impact. By fostering a natural balance between plants, companion planting encourages biodiversity, enhances soil health, and minimizes the need for chemical inputs, making it an effective strategy for organic and environmentally-friendly farming.

Some Companion Planting Schedule

thisunboundlife.com		Basic Companion Planting Chart		thisunboundlife.com	
Grow well together		Beneficial to garden overall	Combo helps pest control	Dont plant together	
	Asparagus, Beets, Marigold, Oregano, Peppers, Radish, Tomatoes, Cucumber, Fennel, Sage, Thyme		Broccoli, Cauliflower, Lettuce, Strawberry, Tomatoes, Sage, Thyme, Beans, Peas		Carrots, Cucumber, Oregano, Parsley, Peas, Squash, Swiss Chard, Tomatoes, Sage, Thyme, Rosemary, Beans, Broccoli, Cucumber
	Broccoli, Carrots, Cauliflower, Cucumber, Peas, Strawberry, Swiss Chard, Tomatoes, Sage, Thyme, Radishes, Cilantro, Chives, Garlic, Leeks, Marigold, Onion, Peppers		Carrots, Onion, Spinach, Sage, Thyme, Beans, Peas		Beans, Cauliflower, Peppers, Sage, Thyme, Broccoli, Carrots, Basil, Cucumbers, Mint, Potatoes, Tomatoes
	Beans, Carrots, Chives, Cucumber, Lettuce, Nasturtium, Onion, Sage, Spinach, Swiss Chard, Dill, Garlic, Onion, Oregano, Rosemary, Thyme, Peppers, Squash, Strawberry, Tomatoes		Carrots, Cucumber, Dill, Spinach, Squash, Strawberry, Tomatoes, Sage, Thyme, Chives, Cilantro, Garlic, Onion, Broccoli, Cabbage, Cauliflower, Kale		Broccoli, Carrots, Cauliflower, Basil, Beans, Chives, Corn Dill, Lettuce, Peas, Peppers, Spinach, Squash, Strawberries, Swiss Chard, Chives, Cucumber, Garlic, Leeks
	Beans, Broccoli, Cauliflower, Leeks, Lettuce, Parsley, Peas, Peppers, Cilantro, Chives, Onion, Rosemary, Sage, Thyme, Tomatoes, Dill		Beans, Broccoli, Cucumbers, Eggplant, Lettuce, Melon, Potatoes, Pumpkins, Tomatoes, Sage, Thyme, Garlic, Leeks, Squash, Beans, Cabbage, Peppers		Broccoli, Cauliflower, Leeks, Lettuce, Peas, Strawberry, Sage, Thyme, Cilantro, Fennel, Potatoes
	Beans, Carrots, Cucumber, Dill, Garlic, Lettuce, Nasturtium, Onion, Sage, Spinach, Swiss Chard, Thyme, Chives, Oregano, Rosemary, Peppers, Squash, Strawberry, Tomatoes		Beans, Peas, Onions, Marigold, Nasturtium, Sage, Thyme, Corn, Peppers, Potatoes, Sunflowers, Squash		Corn, Beans, Lettuce, Melon, Peas, Peppers, Tomatoes, Sage, Thyme, Dill, Nasturtium, Radish, Garlic, Beets, Broccoli, Cauliflower, Fennel, Melons, Onions
	Beets, Cabbage Carrots, Cauliflower, Parsley, Potatoes, Strawberry, Tomatoes, Sage, Thyme, Asparagus, Beans, Spinach, Peas		Broccoli, Cauliflower, Cucumber, Nasturtium, Sage, Thyme, Strawberry, Melon, Beans, Squash, Mint, Sage, Thyme, Tomatoe		Beans, Garlic, Lettuce, Onion, Peas, Spinach, Thyme, Sage, Marigold, Broccoli, Cauliflower, Fennel, Kale
	Beans, Peas, Sage, Spinach, Lavender, Thyme, Rosemary, Fennel		Broccoli, Carrots, Cauliflower, Cucumber, Dill, Leeks, Lettuce, Parsley, Strawberry, Swiss Chard, Tomatoes, Sage, Thyme, Beans, Peas		Corn, Lettuce, Melon, Pumpkin, Squash, Sage, Thyme, Fennel, Swiss Chard, Potatoes
	Beans, Cucumber, Melon, Parsley, Peas, Squash, Sunflower, Sage, Thyme, Dill, Mint, Radish, Broccoli, Cabbage, Tomatoes		Asparagus, Cucumbers, Basil, Peppers, Broccoli, Cauliflower, Sage, Thyme, Basil, Chives, Mint		Beans, Broccoli, Cabbage, Carrots, Cauliflower, Onion, Peppers, Radish, Sage, Thyme, Nasturtium, Marigold, Cucumber, Melons, Mint, Pumpkin, Strawberry, Spinach
	Beans, Broccoli, Cauliflower, Corn, Dill, Lettuce, Onion, Peas, Peppers, Tomatoes, Thyme, Nasturtium, Radish, Potatoes, Sage, Squash		Asparagus, Beans, Broccoli, Cauliflower, Chives, Corn, Onions, Peas, Peppers, Tomatoes, Sage, Thyme, Carrots, Lettuce, Mint		Broccoli, Cauliflower, Strawberry, Beans, Carrots, Chives, Lettuce, Melon, Onion, Peas, Peppers, Spinach, Squash, Strawberries, Swiss Chard, Basil, Celery, Cucumber, Mint, Tomatoes
	Broccoli, Cauliflower, Corn, Cucumber, Lettuce, Onion, Sage, Thyme, Carrots, Tomatoes		Beans, Carrots, Corn, Cucumber, Parsley, Peppers, Spinach, Squash, Strawberry, Sage, Thyme, Chives, Garlic, Leeks, Onion		Basil, Beans, Chives, Cucumber, Garlic, Lettuce, Marigold, Onion, Parsley, Peppers, Thyme, Nasturtium, Marigold, Carrots, Broccoli, Cauliflower, Corn